WORKED SOLUTION

STEP II 2019, Question 10

Mechanics • Rigid Body Equilibrium

STEP II 2019, Question 10

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

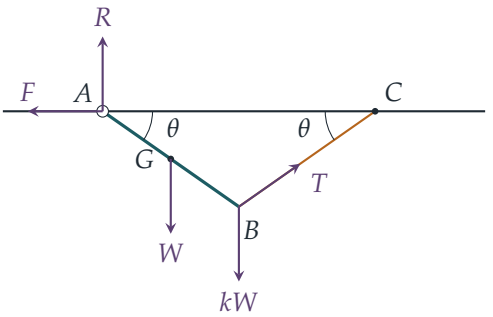
The key idea

Since $AB = BC = 2a$ and A, C both lie on the rail, triangle ABC is isosceles, so the string also makes angle θ with the rail. With that geometry in hand this is a standard rigid-body equilibrium problem: taking moments about A gives the tension without involving the unknown forces at the ring, and resolving then gives the normal reaction and friction. The final part compares two thresholds on k: which one is reached first as k grows?

Solution

Model and forces. The rod is uniform and rigid, so its weight W acts at its midpoint G, with $AG = a$. The ring is light, so the rail's forces on the ring are passed straight on to the rod at A: a normal reaction R vertically upwards and a friction force F along the rail. The string is light and inextensible, with tension T acting at B along BC. The extra load kW acts vertically downwards at B.

In triangle ABC the sides AB and BC are both $2a$, so the base angles at A and C are equal: the string makes angle θ with the rail. Horizontally, B is $2a\cos\theta$ from A, and C is $4a\cos\theta$ from A.



We take F to act along the rail away from C; we shall see that it must, because the string pulls the rod towards C.

Part (i)

Take moments about A, which removes R and F. The weight and the load have moment arms $a\cos\theta$ and $2a\cos\theta$. For the tension, split it at B into $T\cos\theta$ horizontally (towards C) and $T\sin\theta$ vertically upwards. The vertical part has arm $2a\cos\theta$ and the horizontal part has arm $2a\sin\theta$ (the depth of B below the rail), and both turn the rod the same way, anticlockwise in the diagram. So

$$T\sin\theta\cdot 2a\cos\theta + T\cos\theta\cdot 2a\sin\theta = W\cdot a\cos\theta + kW\cdot 2a\cos\theta,$$
$$4aT\sin\theta\cos\theta = aW(1+2k)\cos\theta.$$

Since $\cos\theta \neq 0$ for $0 < \theta < 90^\circ$, we may divide by $a\cos\theta$:

$$T = \frac{(2k+1)W}{4\sin\theta}.$$

This holds whatever the ring does, as long as it does not slip. The string breaks if $T > W$, that is, if $(2k+1)W > 4W\sin\theta$ (multiplying by $4\sin\theta > 0$), which is

$$2k+1 > 4\sin\theta,$$

as required.

Part (ii)

Resolve for the whole rod. Vertically,

$$R + T\sin\theta = W + kW \quad\Longrightarrow\quad R = (1+k)W - \frac{(2k+1)W}{4} = \frac{(2k+3)W}{4}.$$

Horizontally, the only forces are F and the horizontal part of the tension, so

$$F = T\cos\theta = \frac{(2k+1)W\cos\theta}{4\sin\theta}.$$

This is positive, confirming that friction acts away from C. Note also that $R > 0$, so the ring really is pressed onto the rail from above.

Ask yourself When exactly does a ring on a rough rail slip?

Friction can supply at most μR. If equilibrium needs $F > \mu R$, friction cannot hold the ring and it slips. So the ring slips if

$$\frac{(2k+1)W\cos\theta}{4\sin\theta} > \mu\,\frac{(2k+3)W}{4}.$$

Multiplying by $4\sin\theta/(W\cos\theta) > 0$ gives

$$2k+1 > (2k+3)\mu\tan\theta,$$

as required.

Part (iii)

Write $t = \tan\theta$ and $s = \sin\theta$. Each condition says that k has passed a threshold.

The string. By part (i), it breaks once $2k + 1 > 4s$, that is, once

$$k > k_{\text{break}} = \frac{4s - 1}{2}.$$

The ring. Rearranging the condition in part (ii),

$$2k + 1 > (2k + 3)\mu t \iff 2k(1 - \mu t) > 3\mu t - 1.$$

Here we use $\mu t < 1$: dividing by $2(1 - \mu t) > 0$ keeps the inequality the same way round, so the ring slips once

$$k > k_{\text{slip}} = \frac{3\mu t - 1}{2(1 - \mu t)}.$$

(Without $\mu t < 1$ the ring would never slip. If $\mu t \geq 1$ then $(2k + 3)\mu t \geq 2k + 3 > 2k + 1$ for every k. That is why the condition is given.)

As k increases from zero, whichever threshold is lower is crossed first. So the ring slips before the string breaks exactly when $k_{\text{slip}} < k_{\text{break}}$. Multiplying first by $1 - \mu t > 0$,

$$\begin{aligned}
\frac{3\mu t - 1}{1 - \mu t} < 4s - 1 &\iff 3\mu t - 1 < (4s - 1)(1 - \mu t) \\
&\iff 3\mu t - 1 < 4s - 1 - 4s\mu t + \mu t \\
&\iff 2\mu t + 4s\mu t < 4s \\
&\iff \mu t(1 + 2s) < 2s \\
&\iff \mu < \frac{2\sin\theta}{\tan\theta\,(1 + 2\sin\theta)} = \frac{2\cos\theta}{1 + 2\sin\theta},
\end{aligned}$$

dividing by $t(1 + 2s) > 0$ in the last step. This is the required condition.

The rod is in equilibrium to begin with, so at $k = 0$ neither threshold has yet been passed. Both thresholds are crossed only once, because each condition is "k greater than something". So comparing the two thresholds really does decide which happens first.

Checks

- Taking moments about B instead gives the same answer. The forces at A have moment $R \cdot 2a\cos\theta - F \cdot 2a\sin\theta$ about B, and the weight has moment $Wa\cos\theta$. Substituting the values of R and F gives $\frac{1}{2}(2k + 3)Wa\cos\theta - \frac{1}{2}(2k + 1)Wa\cos\theta = Wa\cos\theta$, which balances the weight exactly.
- Take $\theta = 30°$. Then $k_{\text{break}} = \frac{1}{2}$, and the critical friction coefficient is $2\cos 30° / (1 + 2\sin 30°) = \frac{\sqrt{3}}{2} \approx 0.866$, so $\mu t = 0.866 \times 0.577 = \frac{1}{2}$. With this μ, $k_{\text{slip}} = \frac{3/2 - 1}{2 \cdot \frac{1}{2}} = \frac{1}{2}$ as well, so the two thresholds coincide, as they should at the boundary.

- We confirmed by computer that all the forces and moments balance for many random θ and k. We also increased k in small steps for random θ and μ: the ring slipped first exactly when $\mu < 2\cos\theta/(1+2\sin\theta)$.

Common slips

- Forgetting the horizontal part of the tension in the moment about A. It has arm $2a\sin\theta$ and doubles the moment of the tension, which is where the 4 in $4\sin\theta$ comes from.
- Taking $R = (1+k)W$. The string carries part of the weight, so the normal reaction is smaller than the total load.
- Dividing by $1-\mu\tan\theta$ without saying that it is positive. That is exactly what the given condition $\mu\tan\theta < 1$ is for.

Beyond STEP

This is a statically determinate problem: there are three unknowns (T, R, F) and three equilibrium equations for a rigid body in a plane. Part (iii) shows the typical way such a structure fails, by comparing the load at which each constraint gives way. Engineers ask the same question about which member of a structure fails first. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and torque carefully and goes on to rotational dynamics.

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